

Geometry and forces exerted on a bubble and a drop attached to each other

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We consider a bubble of fluid (labelled by the letter b) and a drop of molten metal (label m) floating in a liquid (label l). Before attachment, the bubble has a radius of R_b^0 and the drop of R_m^0 , (fig. 1, top). By attaching to each other (and once equilibrium is reached), an interface forms between them (label i), that can intrude into the bubble (as depicted in fig. 1, bottom) or into the drop. The bubble now has a radius R_b^0 , the drop R_m^0 and the interface between them, R_i . The pressures inside the bubble, drop and bulk liquid are P_b , P_m and P_l , respectively. The surface tensions between the fluids are σ_{bm} , σ_{bl} , and σ_{ml} .

By the Young-Laplace equation

$$P_b - P_l = 2 \frac{\sigma_{bl}}{R_b^0}, \quad (1)$$

$$P_m - P_l = 2 \frac{\sigma_{ml}}{R_m^0}, \quad (2)$$

$$|P_m - P_b| = 2 \frac{\sigma_{bm}}{R_i}. \quad (3)$$

(This assumes that $P_b > P_l$ and $P_m > P_l$, but nothing about the magnitude of P_m relatively to P_b .) By substituting eq. 1 and eq. 2 into eq. 3

$$R_i = \frac{R_m^0 R_b^0 \sigma_{bm}}{|R_b^0 \sigma_{bl} - R_m^0 \sigma_{ml}|} \quad (4)$$

(Note that if $R_b^0 = R_m^0$ and $\sigma_{bl} = \sigma_{ml}$, then $R \rightarrow \infty$. In other words if the bubble and drop have the same size and the surface tensions are equal, then the interface is flat.) The sign of the difference ($R_b^0 \sigma_{bl} - R_m^0 \sigma_{ml}$) tells us whether the interface intrudes into the bubble (if positive) or into the drop (if negative). If its value is zero, the interface is flat.

The contact angles in the bubble, drop and surrounding liquid are θ_b , θ_m and θ_l , respectively (fig. 2). At equilibrium, the surface tensions and the contact

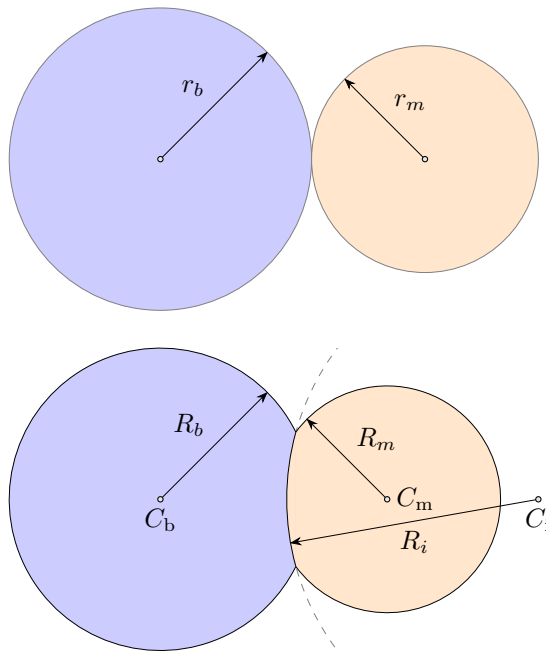


Figure 1: A bubble of gas (blue) and a drop of molten metal (orange) in a liquid, right before attachment (top) and after (bottom).

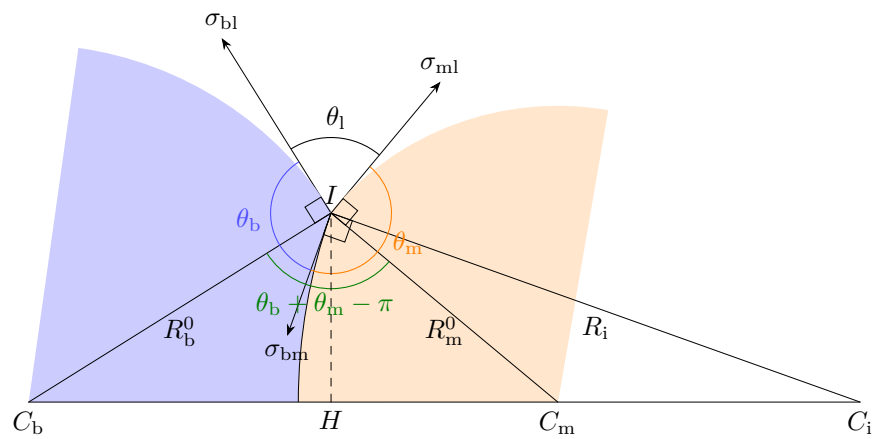


Figure 2: Zoom on the contact between the bubble and the drop

angles are related by

$$\cos \sigma_b = \frac{\sigma_{ml}^2 - \sigma_{bl}^2 - \sigma_{bm}^2}{2\sigma_{bl}\sigma_{bm}}, \quad (5)$$

$$\cos \sigma_m = \frac{\sigma_{bl}^2 - \sigma_{bm}^2 - \sigma_{ml}^2}{2\sigma_{bm}\sigma_{ml}}, \quad (6)$$

$$\cos \sigma_l = \frac{\sigma_{bm}^2 - \sigma_{ml}^2 - \sigma_{bl}^2}{2\sigma_{bl}\sigma_{mm}}. \quad (7)$$

The $\angle C_b I C_m$ angle is equal to $\alpha - \pi$ with

$$\alpha = \theta_b + \theta_m \quad (8)$$

Thus, of the triangle (C_b, I, C_i) we know two sides (R_b^0 and R_i) and one angle. By the law of cosine, the length of the segment $C_b C_m$, noted c , is

$$c \equiv C_b C_m = \sqrt{(R_b^0)^2 + (R_m^0)^2 + 2R_b R_m^0 \cos \alpha}, \quad (9)$$

and the area of the triangle is

$$S = -\frac{1}{2} R_b^0 R_m^0 \sin \alpha. \quad (10)$$

The altitude of the triangle from the vertex I is noted a

$$a \equiv IH = 2\frac{S}{c}. \quad (11)$$

We will need the square of that, that is

$$a^2 = -\frac{1}{2} \frac{(R_b^0)^2 R_m^2 (\sin \alpha)^2}{(R_b^0)^2 + (R_m^0)^2 + 2R_b R_m^0 \cos \alpha}. \quad (12)$$

The generic formula for the heights of the caps of a sphere split in two by a plane is

$$h_1 = r - \sqrt{r^2 - a^2}, \quad (13)$$

$$h_2 = r + \sqrt{r^2 - a^2}, \quad (14)$$

where r is the radius of the sphere and a the radius of the base of the caps ($a \leq r$). The first one h_1 is the height of the smaller of the two caps, h_2 is the height of the larger one. Evidently if $a = r$, then $h_1 = h_2 = r$. The volume of a cap is

$$V^{c,i} = \pi h_i^2 \left(r - \frac{h_i}{3} \right), \quad i \in [1, 2]. \quad (15)$$

and its area

$$A^{c,i} = 2\pi r h_i, \quad i \in [1, 2] \quad (16)$$

So the volume of the metal drop after attachment is the volume of the large spherical cap (on the left of the segment IH on fig. 2) plus or minus the volume of the smaller cap (on the right), depending on whether the interface intrudes into the bubble or into the drop, respectively. The first has a sphere radius of R_m^0 , the second of R_i .

$$V_m = V_m^{c,2} \pm V_i^{c,1}, \quad (17)$$

Conversely, the volume of the bubble is

$$V_b = V_b^{c,2} \mp V_i^{c,1}. \quad (18)$$

The original volumes (before attachment) were

$$V_b^0 = \frac{4}{3}\pi (R_b^0)^3, \quad (19)$$

$$V_m^0 = \frac{4}{3}\pi (R_m^0)^3. \quad (20)$$

The volumes before and after attachment are obviously equal so

$$V_b^0 = V_b^{c,2} - V_i^{c,1}. \quad (21)$$

$$V_m^0 = V_m^{c,2} + V_i^{c,1}, \quad (22)$$

Eqs. 21 and 22 form a system of two non-linear equations from which the whole geometry of the attached bubble and drop can be resolved. In other words, given some initial radii R_b^0 and R_m^0 one should be able to compute the final radii R_b^0 and R_m^0 , and then the rest of the lengths, in particular the distance $C_b C_m$ between the center of the bubble and the center of the drop.

Having done so, we can now focus on the forces exerted on the bubble-drop assembly. The Gibbs free energy of the system before attachment is

$$G^0 = A_b^0 \sigma_{bl} + A_m^0 \sigma_{ml}, \quad (23)$$

where the surface areas are those of spheres

$$A_b^0 = 4\pi (R_b^0)^2, \quad (24)$$

$$A_m^0 = 4\pi (R_m^0)^2. \quad (25)$$

After attachment, the Gibbs free energy of the system is

$$G = A_b \sigma_{bl} + A_m \sigma_{ml} + A_i \sigma_{bm} \quad (26)$$

where the surface areas are those of spherical caps

$$A_b^a = A_b^{c,2}, \quad (27)$$

$$A_m^a = A_m^{c,2}, \quad (28)$$

$$A_i = A_i^{c,1}. \quad (29)$$

The energy of attachment is therefore

$$\Delta G \equiv G - G^0 = (A_b - A_b^0)\sigma_{bl} + (A_m - A_m^0)\sigma_{ml} + A_i\sigma_{bm}, \quad (30)$$

and the force of attachment is defined by

$$F^{\text{att}} \equiv \frac{\Delta G}{\Delta l}, \quad (31)$$

with

$$\Delta l = R_b^0 + R_m^0 - c. \quad (32)$$

Assuming the densities of the liquid metal in the drop, surrounding liquid and fluid in the bubble obey $\rho_m > \rho_l > \rho_b$, the buoyancy force on the bubble (positive, upward) is

$$F_b^{\text{buo}} = V_b(\rho_l - \rho_b)g, \quad (33)$$

on the drop (negative, downward) it is

$$F_m^{\text{buo}} = V_m(\rho_l - \rho_m)g. \quad (34)$$

The net buoyancy force on the bubble-drop system (when they are attached)

$$F^{\text{buo}} = F_b^{\text{buo}} + F_m^{\text{buo}}. \quad (35)$$

Detachment occurs when the positive buoyancy of the bubble and the negative buoyancy of the drop are sufficient to overcome the force of attachment, i.e. when

$$F^{\text{det}} \equiv F_b^{\text{buo}} - F_m^{\text{buo}} > F^{\text{att}}. \quad (36)$$

Note that at zero buoyancy, $F_{\text{buo}}^b = F_m^{\text{buo}}$ and the bubble and drop are necessarily attached.